Reading

Adel S. Sedra and Kenneth C. (KC) Smith, Tony Chan Carusone, and Vincent Gaudet
Microelectronic Circuits 8th Edition, *Oxford University Press*, 2019.

- Chapter 11.2 Some Properties of Negative Feedback

11.3 The Feedback Voltage Amplifier

11.4

Summary

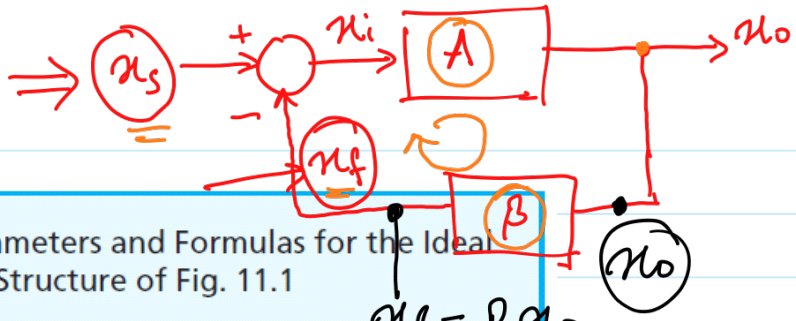
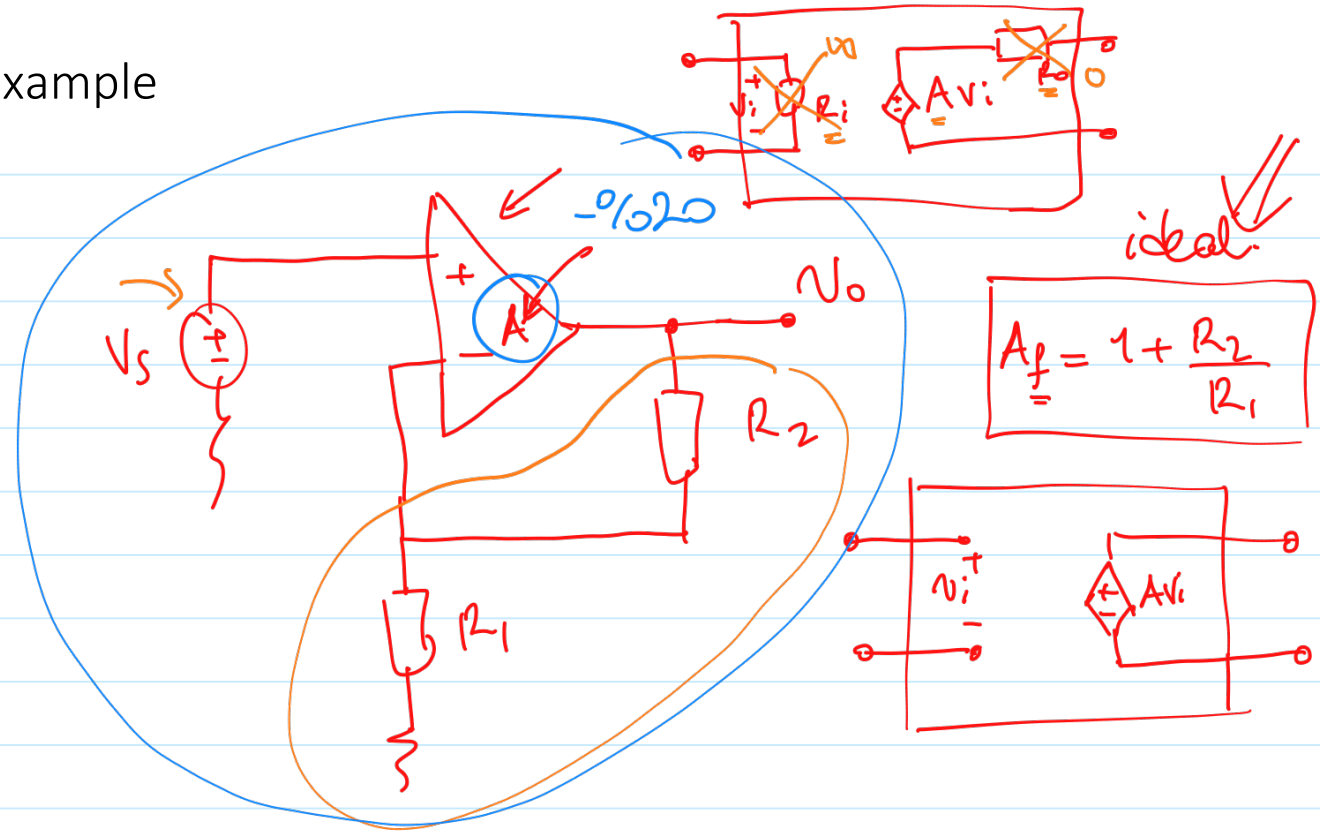


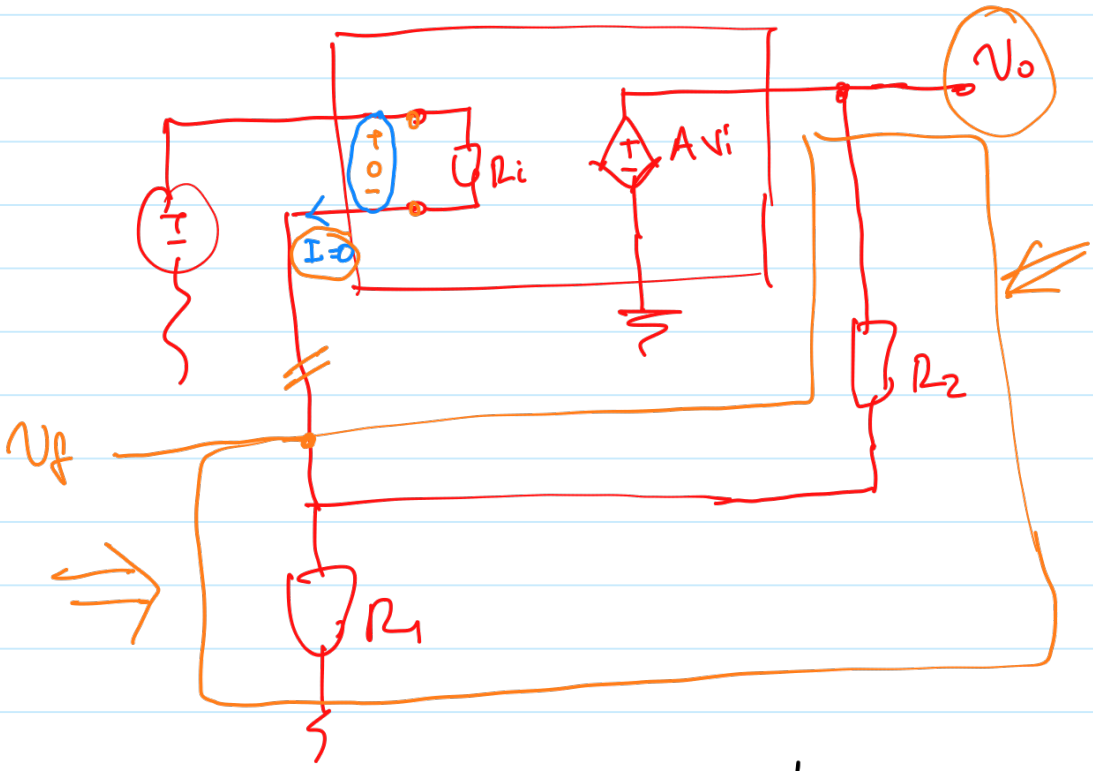
Table 11.1 Summary of the Parameters and Formulas for the Ideal Feedback-Amplifier Structure of Fig. 11.1

- Open-loop gain $\equiv A$
- Feedback factor $\equiv \beta$
- Loop gain $\equiv A\beta$ (positive number)
- Amount of feedback $\equiv 1 + A\beta$
- Closed-loop gain $\equiv A_f = \dfrac{x_o}{x_s} = \dfrac{A}{1 + A\beta}$
- Feedback signal $\equiv x_f = \dfrac{A\beta}{1 + A\beta} x_s$
- Input signal to basic amplifier $\equiv x_i = \dfrac{1}{1 + A\beta} x_s$
- $A_f|_{\text{ideal}} = \dfrac{1}{\beta}$
- Closed-loop gain as a function of the ideal value $\dfrac{1}{\beta}$: $A_f = \left(\dfrac{1}{\beta}\right) \dfrac{1}{1 + 1/A\beta}$
- For $A = \infty$, $x_i = 0$, $x_f = x_s$, $x_o = \dfrac{1}{\beta} x_s$, $A_f = A_f|_{\text{ideal}} = \dfrac{1}{\beta}$
- For large loop gain, $A\beta \gg 1$,

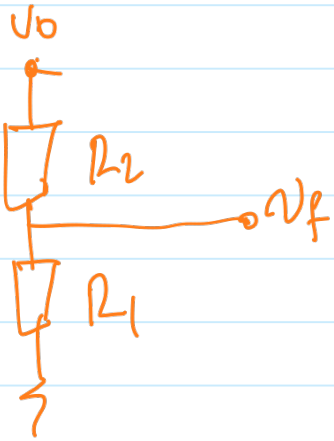
$$A_f \simeq \frac{1}{\beta} \qquad x_f \simeq x_s \qquad x_i \simeq 0$$

Example





$\beta = ?$ $\quad A \to \infty \quad \Rightarrow \quad A_f = \frac{1}{\beta} = \frac{R_1+R_2}{R_1} = 1+\frac{R_2}{R_1}$



$\frac{V_f}{V_o} = \frac{R_1}{R_1+R_2} = \beta$

(2) $A_f = ?$ $\quad A \to \infty$ $\quad A = 10^4$ V/V

$$\beta = \frac{R_1}{R_2 + 9R_1} = 1/10 = 0.1$$

(10)

$$A_f = \frac{A}{1 + A\beta} = \frac{10^4}{1 + 1000} = 9.99 \text{ V/V}$$

(3) $A = 10^4$ V/V $\xrightarrow{-\%20}$ $A = 8000$?

$$A_f = \frac{A}{1 + A\beta} = \frac{8000}{801} = 9,987$$ (9.99)

Gain Desensitivity

$$dA_f = d\left[\frac{A}{1+A\beta}\right] \qquad A \cdot \frac{1}{1+A\beta}$$

$$\frac{dA_f}{A_f} = \frac{dA}{1+A\beta} - \frac{A}{(1+A\beta)^2} = \frac{\frac{dA \cdot (1+A\beta)^2}{(1+A\beta)}}{\frac{A}{1+A\beta}}$$

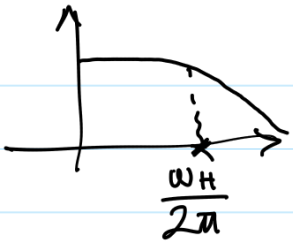
$$\frac{dA_f}{A_f} = \frac{dA}{A} \cdot \frac{1}{(1+A\beta)}$$

$$\frac{\Delta A_f}{A_f} = \frac{\Delta A}{A} \cdot \frac{1}{(1+A\beta)}$$

desensitivity factor

Bandwidth Extension

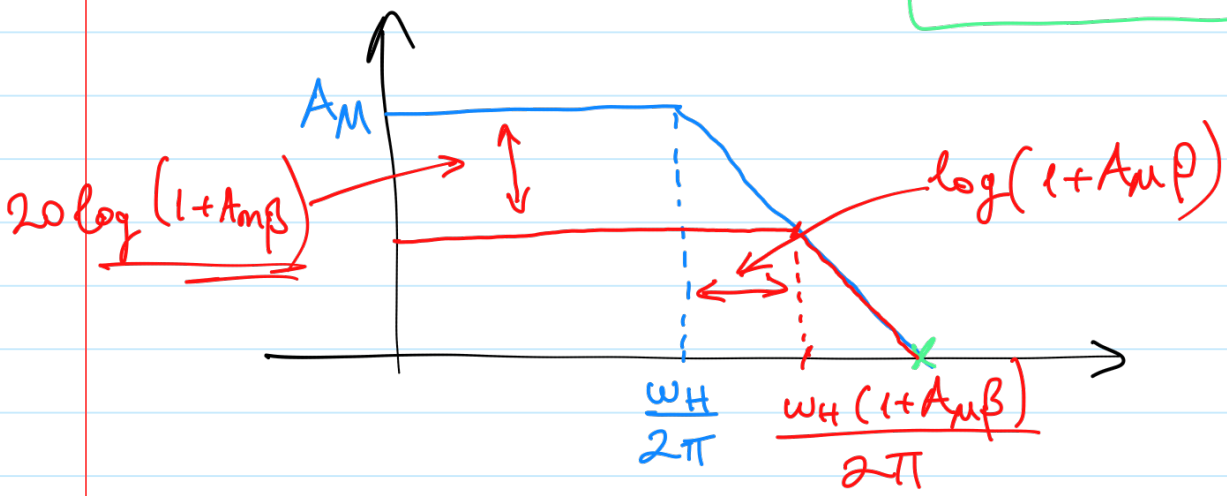
$$A(s) = \frac{A_M}{1 + s/\omega_H}$$



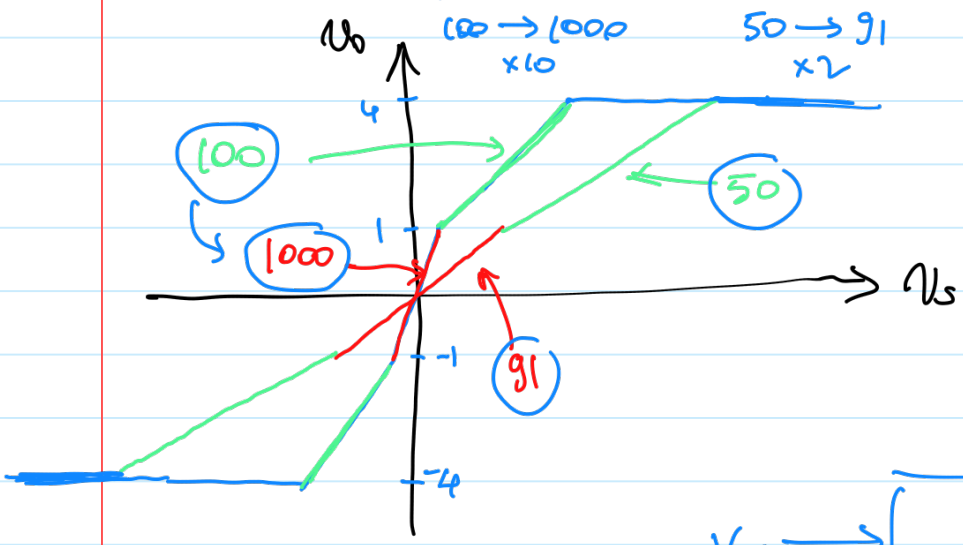
$$A_f(s) = \frac{A(s)}{1 + A(s)\beta}$$

$$A_f(s) = \frac{\frac{A_M}{1 + s/\omega_H}}{1 + \frac{A_M}{1 + s/\omega_H}\cdot\beta}$$

$$= \frac{A_M/(1 + A_M\beta)}{1 + s/[\omega_H(1 + A_M\beta)]}$$

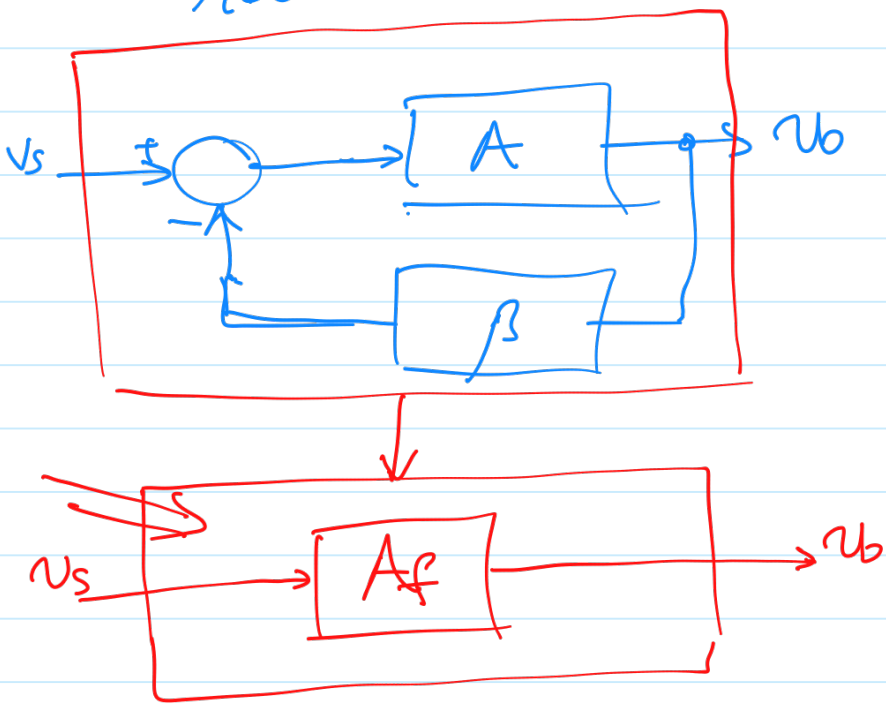


Reduction in Nonlinear Distortion





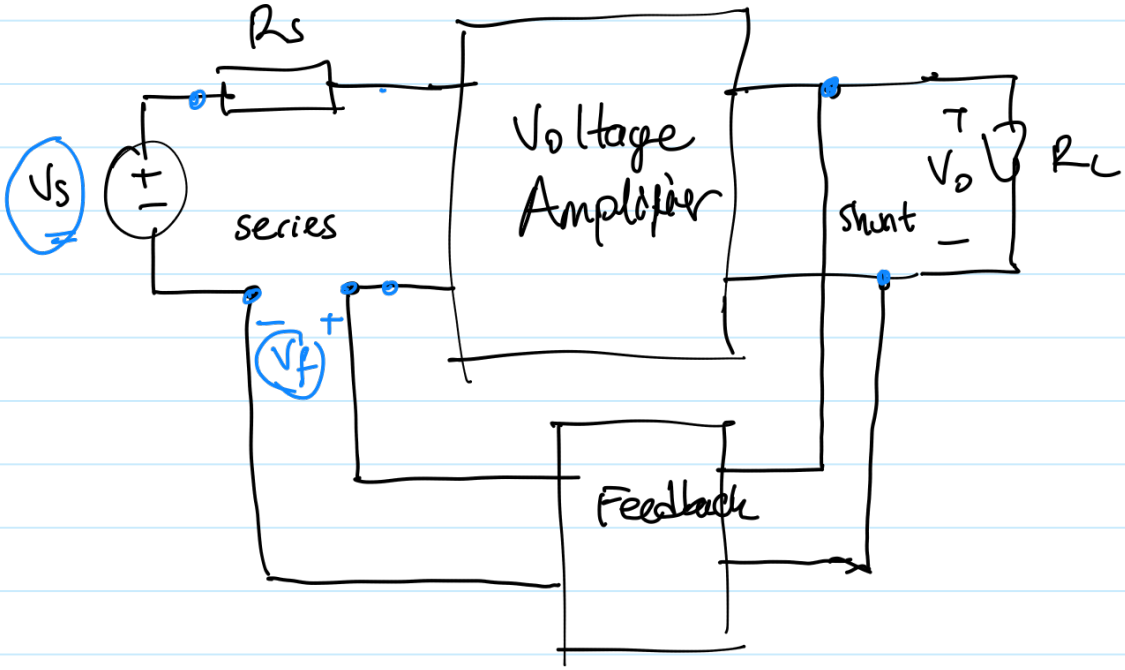
$$\beta = 0.01 = 1/100$$



$$A_f = \frac{A}{1 + A\beta} = \frac{1000}{1 + 10} = 90.9$$

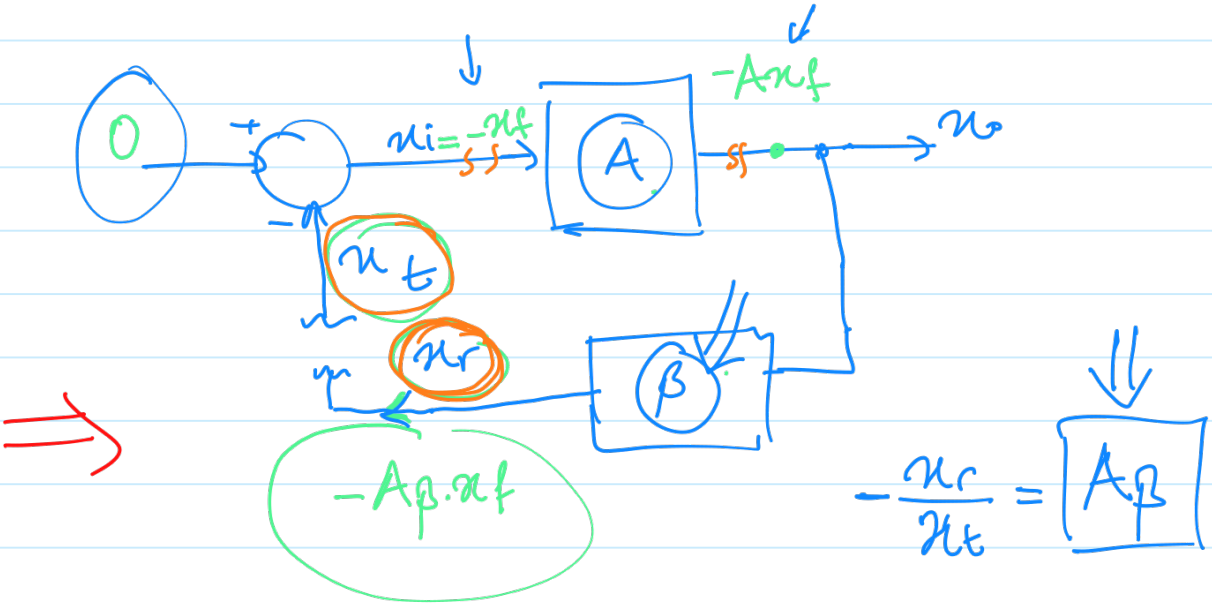
$$A_f = \frac{A}{1 + A\beta} = \frac{100}{1 + 1} = 50$$

The Feedback Voltage Amplifier



Series – shunt

voltage mixing – voltage sampling

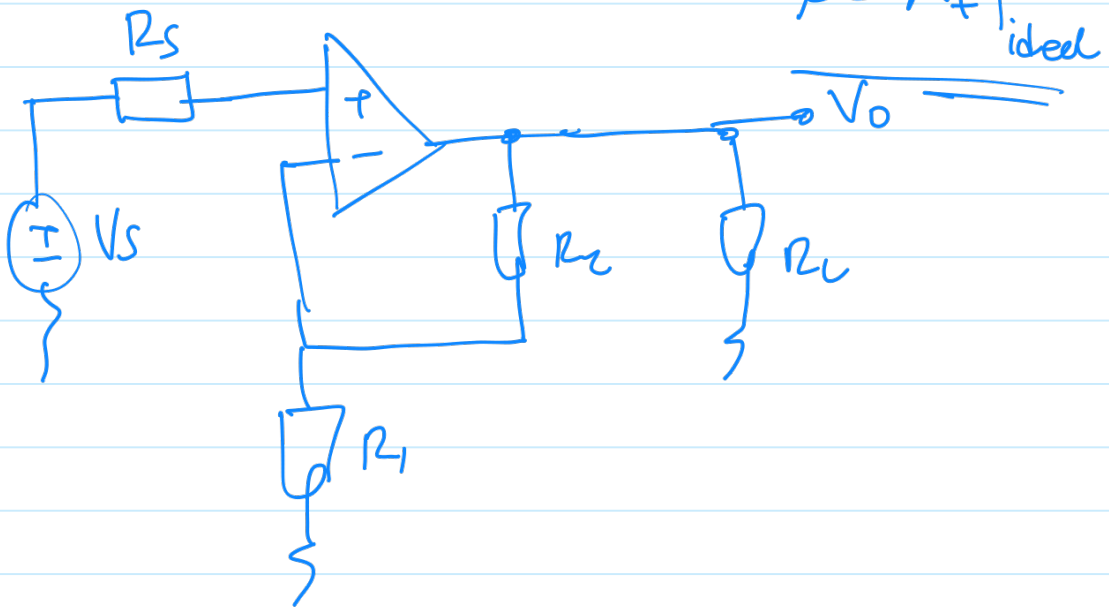


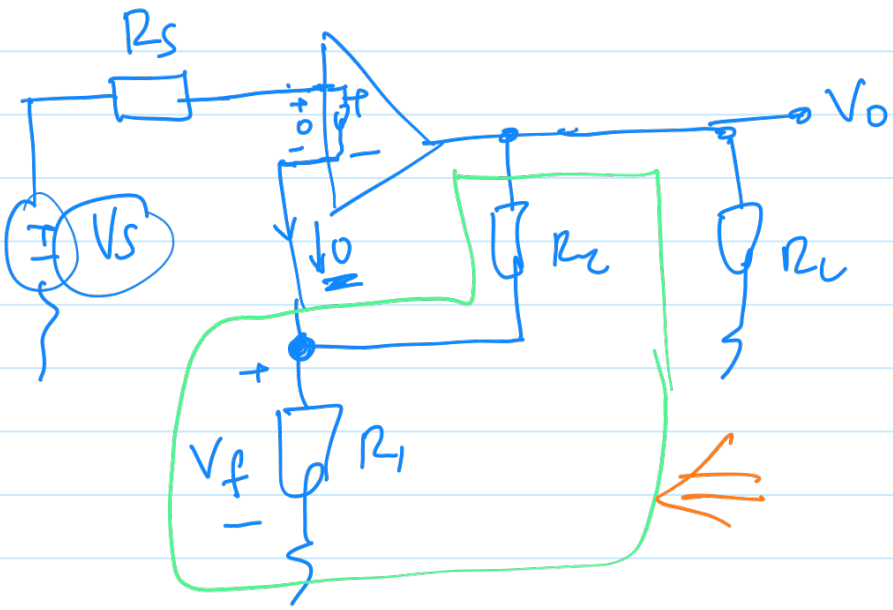
$$\frac{-A\beta \cdot x_f}{x_f} = -A\beta$$

Example - 1

loopgain $A\beta = ?$

$\beta = A_f\big|_{ideal}$

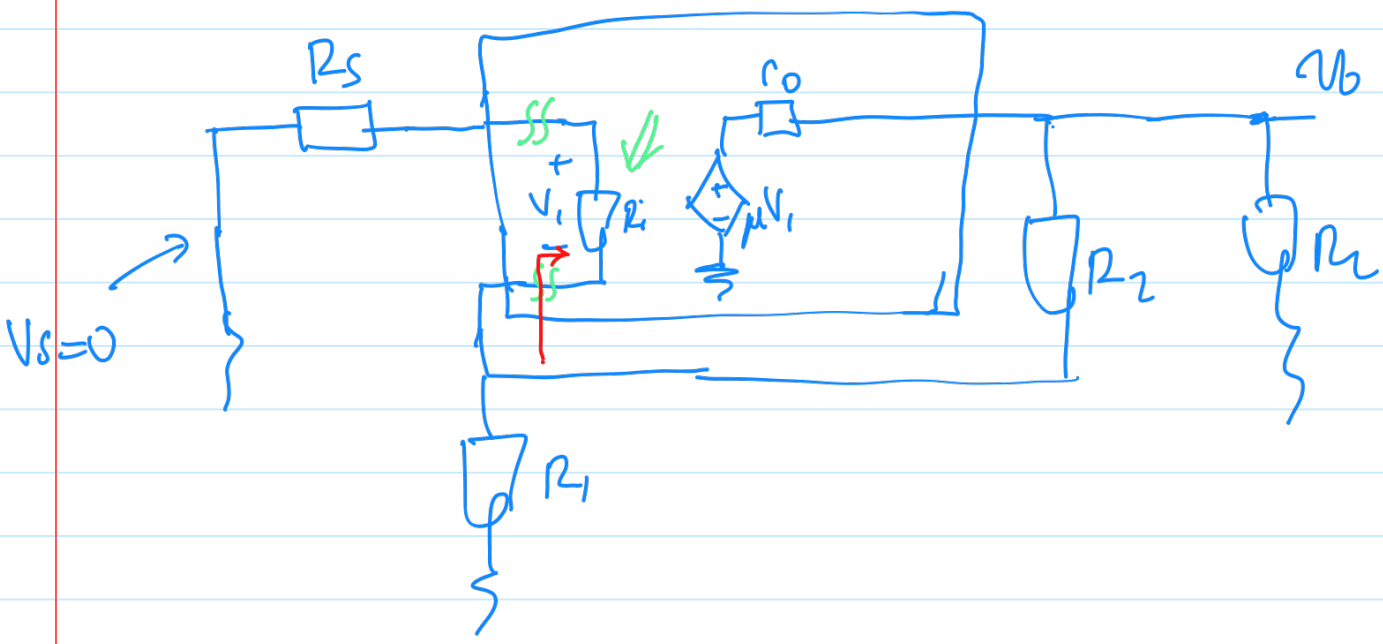


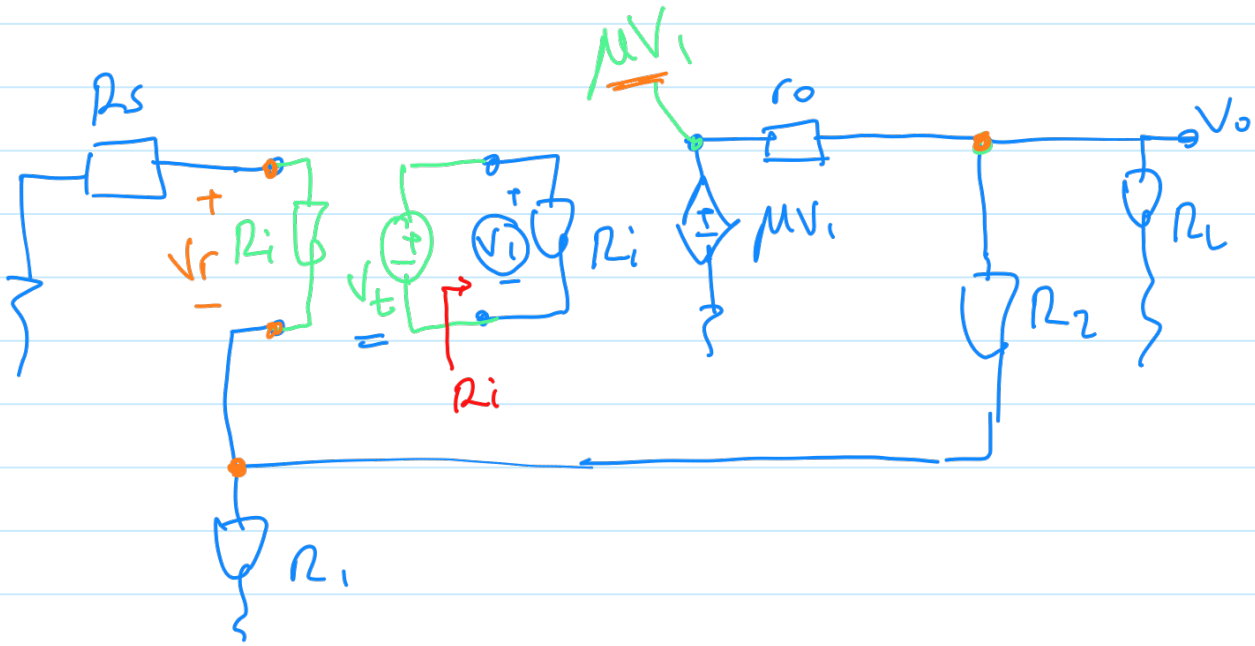


$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2} = 0.0099$$

Example - 1

A_β





$V_t = V_1$

$$\frac{V_r}{V_1} = \mu \cdot \frac{\left((R_s + r_i) \,//\, R_1 + R_2 \right) // R_L}{\left((R_s + r_i) \,//\, R_1 + R_2 \right) // R_L + r_o} \cdot \frac{(R_s + R_i) // R_1}{(R_s + R_i) // R_1 + R_2}$$

$$\cdot \frac{R_i}{R_i + R_s} \Rightarrow A_\beta = 60$$

Example - 1

$A\beta = 60$

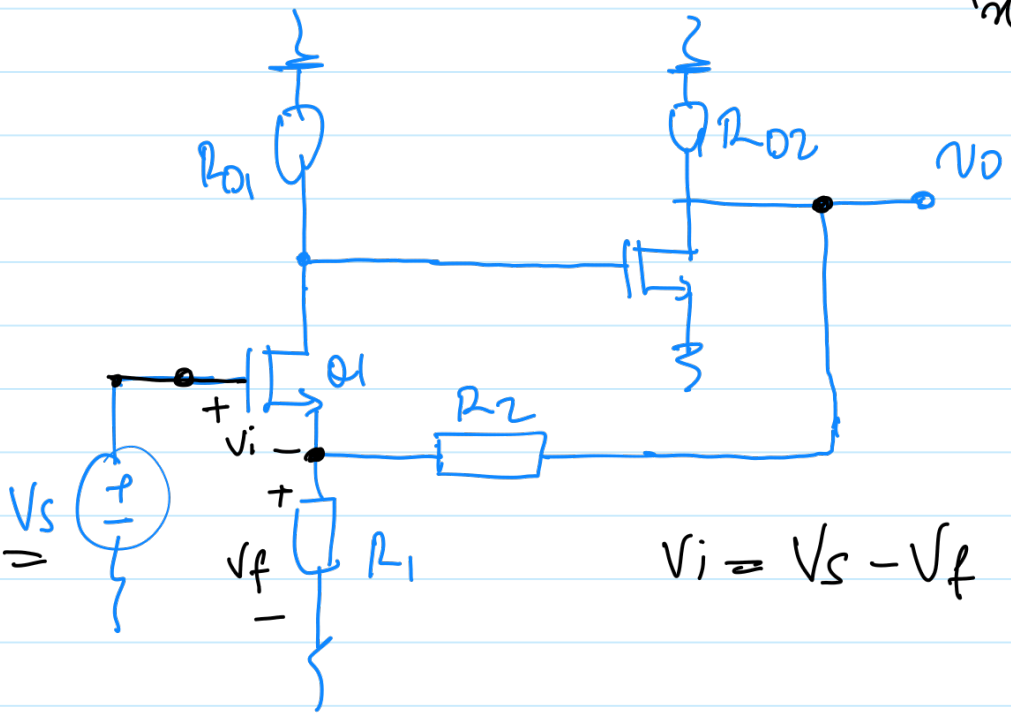
$\beta = 0.0099$

$$\frac{A\beta}{\beta} = A = 5960 \text{ V/V}$$

$$A_f = \frac{A}{1 + A\beta} = 99.3 \text{ V/V}$$

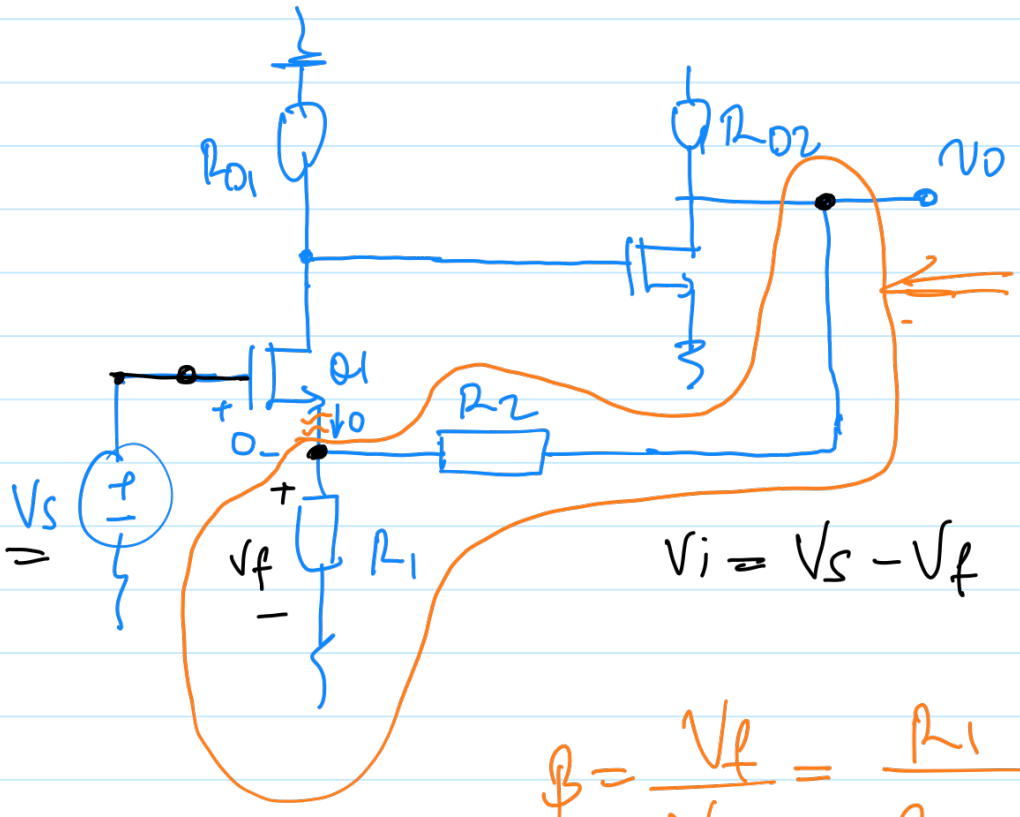
Example - 2





$V_i = V_s - V_f$

$\beta = ?$ $\quad A_f\Big|_{ideal} = ?$ $\quad A \to \infty$ $\quad v_i \to 0$

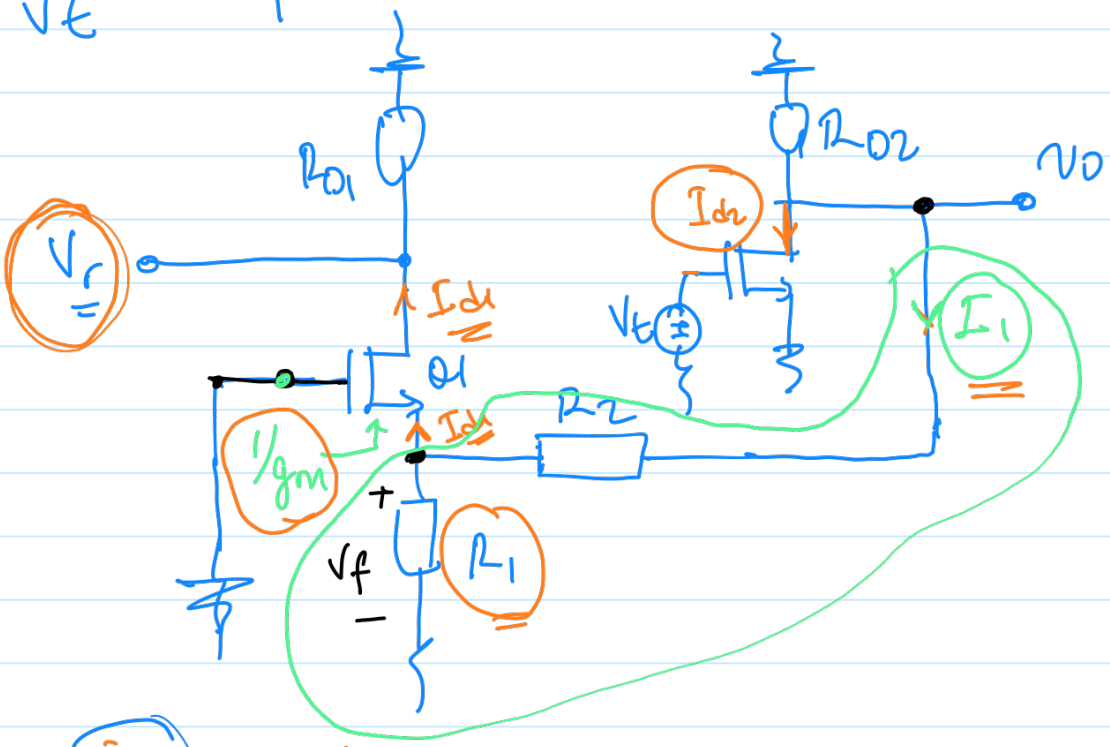


$V_i = V_s - V_f$

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

Example - 2

$$-\frac{V_f}{V_t} = A\beta$$



$$I_{d2} = g_m V_t$$

$$I_1 = -I_{d2}\,\frac{R_{D2}}{(1/g_m \,//\, R_1) + R_2 + R_{D2}}$$

$$I_{d1} = I_1\,\frac{R_1}{1/g_m + R_1}$$

$$V_f = I_{d1} \cdot R_{D1}$$

$$A\beta = (g_{m1}R_{D1})(g_m R_{D2})\,\frac{1}{1+g_{m1}R_1}\,\frac{R_1}{R_{D2}+R_2+(R_1 \,//\, 1/g_{m1})}$$

$$= 16.67.$$

Example - 2

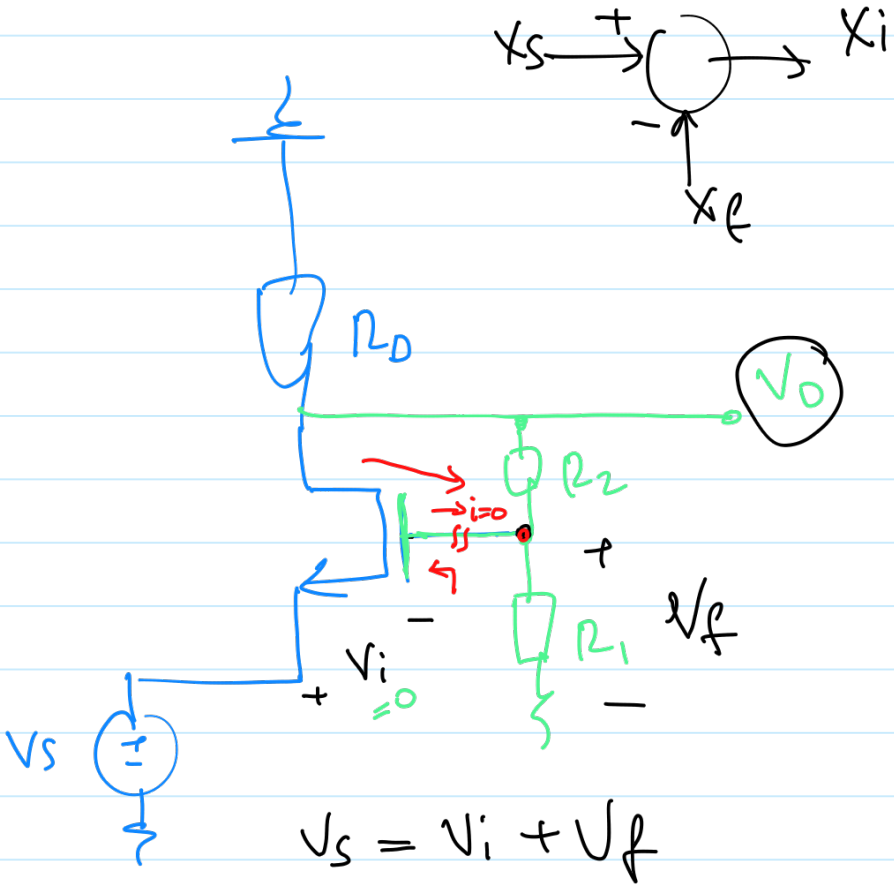
$$\beta = \frac{R_1}{R_1+R_2} = 0.1 \Leftarrow 1/\beta = A_f|_{ideal}$$

$$A\beta = 16.67 \qquad \beta = 0.1$$

$$\frac{A\beta}{\beta} = A = 166.7 \ V/V$$

$$A_f = \frac{A}{1+A\beta} = \frac{166.7}{17.67} = 9.43$$

Example - 3

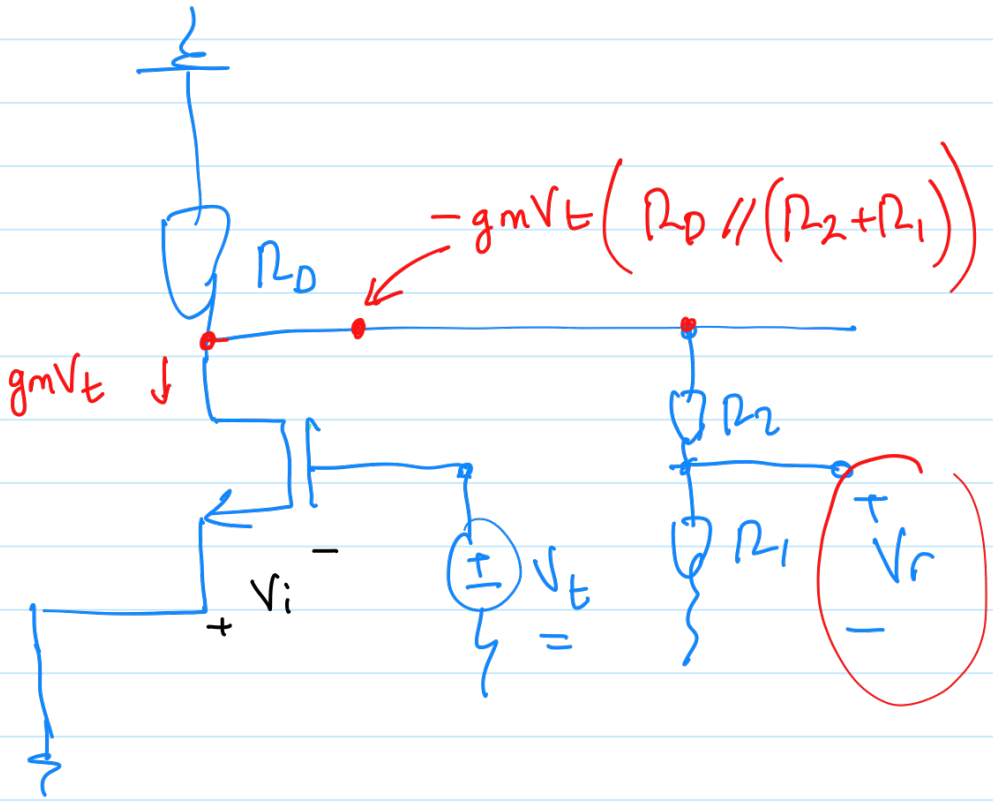


$$V_s = V_i + V_f$$

$$V_s - V_f = V_i$$

$$\beta = \frac{V_f}{V_o} = \frac{R_1}{R_1 + R_2}$$

Example - 3



$$A_\beta = -\frac{V_r}{V_t} = +g_m \left[R_D // (R_2 + R_1) \right] \cdot \frac{R_1}{R_1 + R_2}$$

$$A_f = \frac{A}{1 + A\beta} =$$

Example - 3

Systematic Analysis - Ideal Case

Systematic Analysis - Ideal Case

Systematic Analysis - Practical Case

Systematic Analysis - Practical Case

Template